화학컴퓨터 (2009년 1학기)

기말고사 정답

(**135점**)

1. **(총10점, 각5점)**

(a) Euler 공식을 이용하면,

$$\begin{aligned}(\sqrt{3}+i)^{12} &= \left[2\left\{\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right\}\right]^{12} \\ &= 2^{12}\left\{\cos\left(12\times\frac{\pi}{6}\right)+i\sin\left(12\times\frac{\pi}{6}\right)\right\} \\ &= 2^{12}\left\{\cos\left(2\pi\right)+i\sin\left(2\pi\right)\right\} \\ &= 2^{12}\end{aligned}$$

그리고

$$\begin{aligned}e^{-i\pi/3} &= \cos\left(-\frac{\pi}{3}\right)+i\sin\left(-\frac{\pi}{3}\right) \\ &= \frac{1}{2}-\frac{\sqrt{3}}{2}i\end{aligned}$$

그러므로

$$(\sqrt{3}+i)^{12}\times e^{-i\pi/3} = 2^{12}\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right) = 2^{11}-2^{11}\sqrt{3}i$$

실수부는 2^{11} 이고 허수부는 $-2^{11}\sqrt{3}$ 이다.

(b) $-2+2i = 2^{3/2}\left(-\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}i\right) = 2^{3/2}\big(\cos(3\pi/4)+i\sin(3\pi/4)\big) = 2^{3/2}e^{i3\pi/4}$ 이므로,

$$\begin{aligned}\sqrt[3]{-2+2i}+2e^{i\pi/6} &= \left(2^{3/2}e^{i3\pi/4}\right)^{1/3}+2e^{i\pi/6} \\ &= 2^{1/2}e^{i\pi/4}+2e^{i\pi/3} \\ &= \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right)+2\left(\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right) \\ &= \sqrt{2}\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)+2\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right) \\ &= (1+i)+(\sqrt{3}+i) = 1+\sqrt{3}+2i.\end{aligned}$$

그러므로 실수부분은 $1+\sqrt{3}$ 이고 허수부분은 2 이다.

2. **(총10점, 각5점)**

(a) $\frac{d[A]}{dt} = -k[A]^2$. (단, $t=0$ 일 때 $[A]=A_0$)

$$\begin{aligned}\frac{d[A]}{[A]^2} &= -kdt \\ -\frac{1}{[A]} &= -kt+C \qquad (1)\end{aligned}$$

$t = 0$ 일 때 $[A] = A_0$ 이므로 (1) 식에 대입하면 $C = -\frac{1}{A_0}$ 그러므로 특수해는

$$-\frac{1}{[A]} = -kt - \frac{1}{A_0} \Rightarrow \frac{1}{[A]} = kt + \frac{1}{A_0}$$

(b) $\frac{dm}{dt} + \frac{4m}{20+t} = 2.$ (단, $m(0) = 20$) 적분인자법을 이용하면 위 식은

$$\begin{aligned}
&e^{\int \frac{4}{20+t}dt}\frac{dm}{dt} + e^{\int \frac{4}{20+t}dt}\frac{4m}{20+t} = 2e^{\int \frac{4}{20+t}dt} \\
&\frac{d}{dm}\Big(e^{\ln(20+t)^4}m(t)\Big) = 2e^{\ln(20+t)^4} \\
&\frac{d}{dm}\Big((20+t)^4 m(t)\Big) = 2(20+t)^4 \\
&(20+t)^4 m(t) = \frac{2}{5}(20+t)^5 + C \Rightarrow m(t) = \frac{2}{5}(20+t) + \frac{C}{(20+t)^4}
\end{aligned}$$

$m(0) = 20$ 이므로 (2b) 식에 대입하면 $C = 12 \times 20^4$. 그러므로 특수해는

$$m(t) = \frac{2}{5}(20+t) + \frac{12 \times 20^4}{(20+t)^4}$$

(c) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 - 4 = 0 \Rightarrow (m-2)(m+2) = 0 \Rightarrow m = 2 \text{ 또는 } m = -2$$

$$y(x) = c_1 e^{2x} + c_2 e^{-2x}$$

$y(0) = 2$ 과 $y'(0) = 4$ 을 위 식에 대입하면

$$\begin{aligned}
c_1 + c_2 = 2 \quad &\text{그리고} \quad 2c_1 - 2c_2 = 4 \\
c_1 = 2 \quad &\text{그리고} \quad c_2 = 0 \Rightarrow y(x) = 2e^{2x}
\end{aligned}$$

(d) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3$$

$$y(x) = c_1 e^{3x} + c_2 x e^{3x}$$

$y(0) = 1$ 과 $y'(0) = 0$ 을 위 식에 대입하면

$$\begin{aligned}
c_1 = 1 \quad &\text{그리고} \quad 3c_1 + c_2 = 0 \\
c_1 = 1 \quad &\text{그리고} \quad c_2 = -3 \Rightarrow y(x) = e^{3x} - 3xe^{3x}
\end{aligned}$$

3. **(총 10 점, 각 5 점)**

(a)

$$\begin{aligned}\left[\frac{d^2}{dx^2}, x\right] f &= \frac{d^2}{dx^2}(xf) - x\frac{d^2}{dx^2}f \\ &= \frac{d}{dx}(f + xf') - xf'' \\ &= f' + f' + xf'' - xf'' = 2f' = 2\frac{df}{dx}\end{aligned}$$

$$\left[\frac{d^2}{dx^2}, x\right] = 2\frac{d}{dx}$$

(b)

$$\begin{aligned}\left[\frac{d}{dx} - x, \frac{d}{dx} + x\right] f &= \left(\frac{d}{dx} - x\right)\left(\frac{d}{dx} + x\right) f - \left(\frac{d}{dx} + x\right)\left(\frac{d}{dx} - x\right) f \\ &= \frac{d^2 f}{dx^2} + \frac{d}{dx}(xf) - x\frac{df}{dx} - x^2 f - \frac{d^2 f}{dx^2} + \frac{d}{dx}(xf) - x\frac{df}{dx} + x^2 f \\ &= f'' + f + xf' - xf' - x^2 f - f'' + f + xf' - xf' + x^2 f = 2f\end{aligned}$$

$$\left[\frac{d}{dx} - x, \frac{d}{dx} + x\right] = 2$$

4. **(5점)**

$$\begin{aligned}\mathbf{F} &= -\nabla V(x, y, z) \\ &= -\left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right) V(x, y, z) \\ &= -\mathbf{i}\frac{\partial V}{\partial x} - \mathbf{j}\frac{\partial V}{\partial y} - \mathbf{k}\frac{\partial V}{\partial z} \\ &= -\mathbf{i}k_x x - \mathbf{j}k_y y - \mathbf{k}k_z z.\end{aligned}$$

5. **(5점)**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & 1 \\ 3 & -1 & 1 \end{vmatrix}, \tag{2}$$

$$\begin{aligned}&= \mathbf{i}(1 - (-1)) - \mathbf{j}(-2 - 3) + \mathbf{k}(2 - 3) \\ &= 2\mathbf{i} + 5\mathbf{j} - \mathbf{k}.\end{aligned} \tag{3}$$

6. **(10점)** 비자명한 해를 가지므로 영년식은

$$\begin{vmatrix} -1 - \lambda & 2 \\ 2 & 2 - \lambda \end{vmatrix} = 0$$

$$(-1-\lambda)(2-\lambda)-4=0$$
$$-2-\lambda+\lambda^2-4=0$$
$$\lambda^2-\lambda-6=0$$
$$(\lambda-3)(\lambda+2)=0 \Rightarrow \lambda=3 \quad \text{또는} \quad \lambda=-2$$

(a) $\lambda=3$ 일 때,

$$-4c_1+2c_2=0 \quad \Rightarrow \quad c_2=2c_1$$

정규화조건: $c_1^2+c_2^2=1$ 을 이용하면,

$$c_1=\pm\frac{1}{\sqrt{5}} \quad \text{and} \quad c_2=\pm\frac{2}{\sqrt{5}}.$$

(b) $\lambda=-2$ 일 때,

$$c_1+2c_2=0 \quad \Rightarrow \quad c_2=-\frac{1}{2}c_1$$

정규화조건: $c_1^2+c_2^2=1$ 을 이용하면,

$$c_1=\pm\frac{2}{\sqrt{5}} \quad \text{and} \quad c_2=\mp\frac{1}{\sqrt{5}}.$$

7. **(10점)**

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi(x,y)+\left(\frac{1}{2}k_x x^2+\frac{1}{2}k_y y^2\right)\psi(x,y)=E\psi(x,y)$$
$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,y,z)}{\partial x^2}-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,y,z)}{\partial y^2}+\left(\frac{1}{2}k_x x^2+\frac{1}{2}k_y y^2\right)\psi(x,y)=E\psi(x,y,z)$$

위 미분방정식의 해가 단일 변수로 이루어진 두 함수의 곱이라고 가정하면,

$$\psi(x,y)=X(x)Y(y).$$

(7) 식을 (4) 식에 대입하면 다음과 같은 결과를 얻는다.

$$-\frac{1}{X(x)}\frac{\hbar^2}{2m}\frac{\partial^2 X(x)}{\partial x^2}+\frac{1}{2}k_x x^2-\frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2}+\frac{1}{2}k_y y^2=E$$

$$-\frac{1}{X(x)}\frac{\hbar^2}{2m}\frac{\partial^2 X(x)}{\partial x^2}+\frac{1}{2}k_x x^2=\frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2}-\frac{1}{2}k_y y^2+E\equiv E_x$$

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2}+\frac{1}{2}k_x x^2X(x)=E_xX(x)$$

$$-\frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2}+\frac{1}{2}k_x x^2=E-E_x=E_y$$

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2}+\frac{1}{2}k_y y^2Y(y)=E_yY(y)$$

$$E=E_x+E_y.$$

8. **(5점)**

$$\frac{dU(T)}{dT}=-\frac{Nh\nu}{(e^{h\nu/k_BT}-1)^2}\times-\frac{h\nu}{k_BT^2}e^{h\nu/k_BT}=\frac{N(h\nu)^2}{k_BT^2}\frac{e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2}$$

9. **(총 20점, 각 10점)**

(a)

$$f(x)=\begin{cases}0 & (-L\le x<0)\\ x & (0\le x<L)\end{cases}$$

Fourier 급수로 나타내기 위하여

$$\begin{aligned}
a_n &= \frac{1}{L}\int_{-L}^{L}f(x)\cos\left(\frac{n\pi x}{L}\right)dx,\\
&= \frac{1}{L}\int_{0}^{L}x\cos\left(\frac{n\pi x}{L}\right)dx,\\
&= \frac{1}{L}\left[\frac{L}{n\pi}x\sin\left(\frac{n\pi x}{L}\right)\right]_0^L-\frac{1}{L}\int_0^L\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)dx \quad \text{부분적분 이용}\\
&= \frac{1}{n\pi}\left[\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_0^L\\
&= \frac{L}{(n\pi)^2}\big(\cos(n\pi)-1\big)\\
&= \frac{L}{(n\pi)^2}\big((-1)^n-1\big)
\end{aligned}$$

$$
\begin{aligned}
b_n &= \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx, \\
&= \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)dx, \\
&= \frac{1}{L}\left[-\frac{L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)\right]_0^L + \frac{1}{L}\int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)dx \quad \text{부분적분 이용} \\
&= -\frac{L}{n\pi}\cos(n\pi) + \frac{1}{n\pi}\left[\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\right]_0^L \\
&= -\frac{L}{n\pi}(-1)^n \\
&= \frac{L}{n\pi}(-1)^{n+1}.
\end{aligned}
$$

한편,

$$
\begin{aligned}
a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)dx \\
&= \frac{1}{L}\int_0^L x dx \\
&= \frac{1}{L}\left[\frac{1}{2}x^2\right]_0^L x dx \\
&= \frac{L}{2}
\end{aligned}
$$

Fourier 급수는 다음과 같이 주어진다.

$$
f(x) = \frac{L}{4} + \sum_{n=1}^{\infty}\frac{L}{(n\pi)^2}\left((-1)^n - 1\right)\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty}\frac{L}{n\pi}(-1)^{n+1}\sin\left(\frac{n\pi x}{L}\right),
$$

(b) $x = 0$을 (9a) 식에 대입하면

$$
\begin{aligned}
f(0) = 0 &= \frac{L}{4} - \frac{2L}{\pi^2} - \frac{2L}{(3\pi)^2} - \frac{2L}{(5\pi)^2} - \frac{2L}{(7\pi)^2} - \cdots \\
&= \frac{L}{4} - \frac{2L}{\pi^2}\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots\right)
\end{aligned}
$$

그러므로

$$
1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots = \frac{\pi^2}{8}
$$

10. **(총 20점, 각 10점)**

(a)

$$\begin{aligned}\frac{dU_{LJ}(r)}{dr} &= 4\epsilon\left[-12\left(\frac{\sigma^{12}}{r^{13}}\right)+6\left(\frac{\sigma^{6}}{r^{7}}\right)\right] \\ &= 24\epsilon\left(\frac{\sigma^{6}}{r^{7}}\right)\left[-2\left(\frac{\sigma^{6}}{r^{6}}\right)+1\right] \\ \frac{dU(r)}{dr}=0 &\Rightarrow -2\left(\frac{\sigma^{6}}{r^{6}}\right)+1=0 \Rightarrow r_0=2^{1/6}\sigma\end{aligned}$$

(b)

$$\begin{aligned}\frac{d^2U_{LJ}(r)}{dr^2} &= \frac{d}{dr}\left\{24\epsilon\left(\frac{\sigma^{6}}{r^{7}}\right)\left[-2\left(\frac{\sigma^{6}}{r^{6}}\right)+1\right]\right\} \\ &= -168\epsilon\left(\frac{\sigma^{6}}{r^{8}}\right)\left[-2\left(\frac{\sigma^{6}}{r^{6}}\right)+1\right]+288\epsilon\left(\frac{\sigma^{6}}{r^{7}}\right)\left(\frac{\sigma^{6}}{r^{7}}\right) \\ &= 624\epsilon\left(\frac{\sigma^{12}}{r^{14}}\right)-168\epsilon\left(\frac{\sigma^{6}}{r^{8}}\right)\end{aligned}$$

$r=2^{1/6}\sigma$ 을 위 식에 대입하면

$$\begin{aligned}\frac{d^2U_{LJ}(r)}{dr^2}\Big|_{r=2^{1/6}\sigma} &= 624\epsilon\left(\frac{\sigma^{12}}{2^{14/6}\sigma^{14}}\right)-168\epsilon\left(\frac{\sigma^{6}}{2^{8/6}\sigma^{8}}\right)=156\epsilon\left(\frac{1}{2^{1/3}\sigma^{2}}\right)-84\epsilon\left(\frac{1}{2^{1/3}\sigma^{2}}\right) \\ &= 72\epsilon\left(\frac{1}{2^{1/3}\sigma^{2}}\right)>0\end{aligned}$$

$r=r_0$ 에서 최소값을 가진다.

11. **(총 10점, 각 5점)**

(a) 질량보존법칙에 의해 $[A]_t+[B]_t=[A]_0+[B]_0$ 이므로 $[B]_t=-[A]_t+[A]_0+[B]_0$ 이고

$$\begin{aligned}\frac{d[A]_t}{dt} &= -k_1[A]_t+k_2[B]_t=-k_1[A]_t+k_2(-[A]_t+[A]_0+[B]_0) \\ &= -(k_1+k_2)[A]_t+k_2([A]_0+[B]_0)\end{aligned}$$

(b) (a) 식을 다시 쓰고 적분인자법을 이용하면

$$\begin{aligned}&\frac{d[A]_t}{dt}+(k_1+k_2)[A]_t=k_2([A]_0+[B]_0) \\ &e^{\int(k_1+k_2)dt}\frac{d[A]_t}{dt}+e^{\int(k_1+k_2)dt}(k_1+k_2)[A]_t=e^{\int(k_1+k_2)dt}k_2([A]_0+[B]_0) \\ &\frac{d}{dt}\left[e^{(k_1+k_2)t}[A]_t\right]=e^{(k_1+k_2)t}k_2([A]_0+[B]_0) \\ &e^{(k_1+k_2)t}[A]_t=\frac{k_2}{k_1+k_2}e^{(k_1+k_2)t}([A]_0+[B]_0)+C \\ &[A]_t=\frac{k_2}{k_1+k_2}([A]_0+[B]_0)+Ce^{-(k_1+k_2)t}\end{aligned}$$

$t = 0$ 에서 $[A]_t = [A]_0$ 이므로 이를 위 식에 대입하면,

$$[A]_0 = \frac{k_2}{k_1 + k_2}([A]_0 + [B]_0) + C \Rightarrow C = \frac{k_1[A]_0 - k_2[B]_0}{k_1 + k_2}$$

그러므로

$$[A]_t = \frac{k_2}{k_1 + k_2}([A]_0 + [B]_0) + \frac{k_1[A]_0 - k_2[B]_0}{k_1 + k_2} e^{-(k_1+k_2)t}$$